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Electronic Delivery of Financial Services

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2 Introduction

Basic economic functions can be identified as the answers to a set of questions -- what to produce, how to produce it, and how to distribute the output. This functional perspective to banking has given rise to the separation of the economic activities of banks into classes like commercial, corporate, small business, or retail. Involving direct contact with customers, retail banking needs the strongest functionalities on its distribution channels. Basic conditions for these channels to succeed can be given as :

1. Strong bank branding.
2. Unique value to customers.
3. Customer centric -- reflecting the customer relationship.
4. Must be easy to use and intuitive to the customer.
5. Finally, and most important, it must be secure!

The innovations in computing and telcommunications have improved all the functions in retail banking. Computers have empowered the production and telecommunications have created new distribution channels that have instantaneous and global reach. Of these new channels, internet is the least costly and with widest reach. However, each channel has its own virtues and for the best benefit of a retail bank, they should co-exist with a highly computerized strong back-end and a middleware to glue them all together.

3 Retail Banking

Using the functional perspective, a retail bank (or the business of "retail banking") is a particular combination of functions performing the transfer of purchasing power across time and distance. Consider a simple example: the savings account. This financial product transfers purchasing power from the present to the future. But this savings function can be accomplished in other ways; for example, by investing in common stock. There are risk differences between the instruments: the savings account comes with a guarantee of principal, and in that regard is safer than stock. The functions provided by the savings account remain the same (transferring purchasing power into the future, reducing risk), but they need not be provided by a depository institution. This multiplicity is changing the nature of competition in retail banking markets.

Retail banks can serve their customers by bundling certain financial, or they can merge banking-like services with non-banking-like services, such as tickets to concerts and sporting events, and vacation planning. These firms may have electronic delivery vehicles and be accessed through the Internet. Prosperous firms will be those that would find ways to deliver services that the public would need and enjoy as an almost shopping like experience.

4 Electronic Delivery Channels and Products

4.1 Alternative Channels

Alternative electronic delivery channels (costing between 10 and 25 cents per transaction compared with $1.50 to $2.50 for transactions involving bank staff) provide a major means to cut costs.

The first renaissance in electronic delivery began in 1969 when Chemical Bank placed a cash dispenser at a branch in Queens, in New York. What followed was the proliferation of ATMs, plastic cards, PINs and a totally revised customer value system. The first renaissance was profound but took about 20 years to fully take hold. The second renaissance is driven by the Internet and is now in the early stages. The most striking difference between the two periods is the financial customer. In the '70s and '80s, financial customers were being led by the financial industry. Today, financial customers are waiting for the industry to catch up to their expectation levels.

Branch	2.95 USD
Call-Center	0.56 USD
Internet	0.04 USD
ATM	0.01 USD

Table 1 : Average cost of a transaction committed on various channels

4.1.1 ATM

As the first replacement of a human teller in banking industry, ATM was a hard product for advertising departments to develop public acceptance. They even used catchy names for these new machines in the hopes that they would be thought of as more human by their customers:. Tillie the Teller, The Bread Box, The Green Machine, Anne the Anytime Teller, Zippy, Millie, The Money Tree, The Cash Cow, etc. Finally, The American Banker declared that they would hereafter refer to the full-function machine as an automated teller machine" or "ATM". Responses to ATMs were mixed even in the bank industry. A bank manager is popularly quoted to say : "Let me get this straight. You want me to poke a hole in the wall of my branch. Then you want me to stick a one-ton machine through that wall. Then that machine is going to spit money out onto the sidewalk. Do I have it right so far?" Another anecdote from 70's is the "Seven Reasons Why ATMs Will Never Catch On :"

1. Customers will never remember a 4 or 6-digit number.
2. Most people do not believe it is right to have such easy access to their money.
3. Not enough customers will carry the extra plastic card in their wallets.
4. There is no demand for 24-hour availability.
5. You can't ask customers to learn to use a machine
6. Customers will reject ATMs because they cannot record the transaction in their check register.
7. Customers will use their cards at the point of sale, and obviate the need for cash.

For one thing, these reasons did not turn out to be completely wrong. Even today, for a retail bank, an average of 30% of consumers still do not use ATMs.

4.1.2 Telephony

Banks and other financial businesses implement call centers for one simple reason: profits. Call centers automate the most basic exchanges, like requesting an account balance or making a deposit, so that agents can devote more time to making sales or providing customers with more personalized service. Moreover, with lower facility costs, a transaction fulfilled by a live agent over the phone can be as low as 1/6th of that of the branch agent. Call centers also help to secure customers' loyalty, ensuring that their individual needs are met by an automated system or a live agent over the phone as quickly as possible. Companies in the financial industry are collectively spending more money to improve their call centers with additional staff and with new hardware and software. Datamonitor, a research and management consulting firm, estimates that the number of agents working at call centers for banks and for retail financial businesses in US, will rise from an estimated 183,000 agents in 1998 to 215,000 by 2003. Datamonitor estimates that labor accounts for 64% of the cost of running a call center, leaving managers with only the remaining 36% to work with when purchasing new equipment and meeting any additional overhead costs. Datamonitor, reports that banks have spent a total of $835 million on call center products and services by the end of 1999. It is predicted that they will spend $1.07 billion this year and $1.35 billion in 2001; parallel trends in other financial services sectors like insurance and brokerage firms.

As financial businesses expand their call centers, they must work within their budgets to create the most reliable systems possible. There is a number of strategic issues in designing a call center :

1. <u>Site selection</u> : A center of large size may not necessarily find enough qualified labor in one location alone and it would also be expensive to operate. Therefore a number of centers can be maintained and the co-working or routing calls can be accomplished by intelligent networking. Prudential, a multi-faceted financial firm based in Newark, NJ that offers investment, securities, insurance and real estate services, established a network of 20 call centers to accommodate its expansive customer base in the US and currently employs more than 4,000 agents, who handled approximately 20 million inbound and outbound calls in 1999. Training and maintaining a high performer agent is not easy. Centers can also

employ remote agents who use telecommuting software that routes calls and tracks them from the call center's server.

2. The office : Office layout is carefully designed to ensure that agents don't feel isolated. Agent psychology is an immediate concern for the center. Agents are provided with all means to achieve higher performance. They use ergonomic desks and seating. They wear cordless headsets, with possibly noise cancellation capabilities. Still, for an average call center, agent turnover may reach upto %40. Since they are trained usually about a month, turnovers are very undesirable for the center.
3. Dialog flow scenarios **:** With the huge volume of calls that centers handle in a day, every second saved, every dialog step skipped or covered by IVR, returns as hours of savings in terms of agent labor.
4. Interactive Voice Response (IVR) : These systems are used to eliminate the need for answering and routing certain types of calls. For basic services like finding out what the customer's balance is an IVR provides a self service functionality. In general IVR can be utilized to manage upto 80% of investment calls, 60% of securities calls, and 25% of retail banking calls.
5. Computer Telephony Interface(CTI) : This is the unit which initiates database queries or transactions by calling functions on the back-end systems. It controls the dialogs and generates onscreen scripts that prompt agents to perform all the necessary tasks such as asking questions to help them identify customers, input account information in the right order etc.
6. Continuous Development **:** Today, call center technologies is a fast developing technology arena. Developers offer numerous highly technical products to increase agent performance. Since the return of investment is almost immediate, it is not at all hard to sell products. Some popular examples are :
   - SmartRoute : Identifies callers by capturing the phone numbers they are calling from before routing calls to agents.
   - Sixth Sense : Receives data from the IVR system and the customer database, to display screen pops on agents' PCs with customers' histories and other relevant information.
   - Voice over IP : Makes use of internet as a free medium for speech communications to drop the cost.
7. Evolving from a call center into a contact center**:** The latest trend that call centers are following to ensure rapid customer service is to route and answer e-mail messages along with phone calls. Agents can answer an average of 200 to 300 e-mail messages a day or as many as 1,000 during busy periods. Call centers operate with less agents at night shifts. Customers who send e-mail during off-hours receive an automated response to confirm that their messages arrived at the call center successfully and that agents will help them by the next business day. An example product, Genesys Internet Suite, automatically responds to customer e-mail or suggests responses. It also enables agents to engage in live on-line chat sessions with customers and speak with them through voice over IP.
8. Mergers and acquisitions **:** Technical innovations are not the only factors that affect call center development for the financial sector. Mergers have been just as influential, particularly among banks. Functionally, the technical infrastructure of a call center can be utilized to serve other banks or even companies from other industries. Adding a new set of dialog scenarios require only agent training to upscale a call centers to work for an additional company.

4.1.3 Internet

Internet banking belongs to the species of financial services variously known as home banking, remote electronic banking, online banking, self-service banking, and other names indicating that customers do their banking at home or at work. Over 1,200 European financial institutions offer Internet banking today, more than twice as many as six months ago. In fact internet turns out to be the most economic delivery channel

for banks. So much so that in Sweden, where banks were the first to deliver internet access service, people consider banks their portals and this enabled them to capture market share before the portals became big. According to a study released in September 2000 by KPMG International, Financial institutions currently derive 13% of their revenues from business done on the Internet, and that share is expected to jump to 28% in the next 18 months.

The world's first Internet bank opened its virtual doors in the fall of 1995. Today, all banks in USA and %80 of banks in Europe offer Internet based banking. Wells Fargo, a US bank who is a very aggressive Internet player, claims 450,000 online customers as of Spring2000, up 50% from the start of last year.

Advantages of online banking to the customers can be listed as follows:

- Convenient
- Inexpensive
- Convenient bill paying medium
- 7/24 availability
- Bank from anywhere in the world (as long as you have access to a computer and the network)

Few disadvantages are :

- Requirement for basic computer and browser literacy.
- Need for a computer with network connection
- Security concerns

One of the two main problems, the computer literacy and internet access, is eroding away as more and more people are starting to use internet everyday. A recent Gallup/American Banker survey found that Over 68% of U.S. households have access to the internet and 63% of U.S. households now own personal computers, but only 12% use them for banking transactions. American Banker concludes that "if not for the fact that PC ownership is expanding, there would be no growth in online banking customers." Nevertheless, PC ownership is growing. By 2003, Europe is projected to have 170 million Internet users, about the number forecast for the United States. Moreover, banking services over the Net is getting mature. Perhaps equally telling in the Gallup survey, 69% of online customers said they were "very satisfied" with the service this year, up from 59% in 1999.

A major promise by online banking is the global reach. U.S. financial players are already seeking to expand their presence in Europe. In addition, there is likely to be a proliferation of pan-European internet banks. The quest to build an on-line European banking presence will no doubt be complicated by current laws and standards, which differ from country to country. In many instances, the pioneers in Europe make their applications to the regulators on a case-by-case basis, taking their time to expand. An example is Scandinavia's largest asset management bank, SEB, which is striving to play a larger role across Europe via the Internet. SEB, with 305,000 on-line customers as of 2Q2000 in the Nordic region, hopes to have five million customers banking via the Internet by 2004. It offers a full range of services that include banking, brokerage, mutual funds, loans, mortgages, and bill pay and presentment.

4.1.4 Mobile phones

Before the advent of internet access using mobile phones, mobile banking was already possible by simply calling the bank call center the mobile phone. Then, what does the wireless internet access provide?

Once again the answer is "cost reduction". The savings are both for the customer and for the bank :

- Customer side **:** In the case of fixed line telephony access to the bank, call charges are low enough to be subsidized by the bank. However, mobile call charges are shared not only because they are very high

but also banks do not want to share customer information with the privately owned mobile operators. Operators with all their calls routed over a few Mobile services Switching Centers (MSC) can dig into the digital data and produce detailed call statistics.

- Bank side **:** Customers who use the mobile browser on the phone to commit banking are essentially doing a self-service. This is a number of times cheaper for the bank, then transactions committed by live agents over the phone.

Establishing the business case for mobile banking, the industry has to make sure they are building a trusted environment where the customer is owned by the bank. At the moment, wireless data services are technically not able to achieve end-to-end security. Secure connection from the handset to the bank is breached at least at one point, when the mobile data packets are converted into internet data packets at the mobile operator premises. Customers are billed by the operator for using the wireless data service hence the general conception is that "the customer is using a mobile banking service brought by the mobile operator". Banking industry, whose core business depends on trust build on direct communication with the customer, do not favor this notion. Therefore banks have to develop the business case for mobile internet banking and broking very carefully.

One major drive which effects the compromise of cost versus business concerns of the banks is the increasing number of wireless devices which provide internet access. New research has found that the number of mobile data users will total nearly 1.2 billion worldwide in 2005. This will exceed users of fixed Internet, which will grow to around 750 million during the same period. Western Europe will lead the way in this rapidly developing market with over 400 million people using data in this region in 5 years time. Strategis Group, estimates that by 2005, European wireless voice revenue will be 155 billion$/year while just location based mobile data services will generate 35 billion$/yr.

	2000	2003	2005
USA	20	120	224
Japan	26	79	113
Asia/Pacific	29	151	282
Western Europe	101	324	409
Rest of the World	7	64	159
World Total	184	737	1,187

Table 2 : Millions of people with mobile internet access

This explosive growth in mobile data will see wireless data communications eventually outpacing voice as the mobile market's driving factor, with 70% of mobile subscribers worldwide using data in 2005.

Despite the business concerns, the big and growing size of the wireless internet market, pushes the players from all industries to seek new service offerings beyond their traditional industry roles and establish partnerships in order to maximize value creation and revenue generation.

Personalization of content and the optimization of the mobility, time and location sensitive features will also prove crucial in the relevance and take up of wireless data services.

4.1.5 TV banking

Together with mobile phones, TV is the latest one of the emerging digital content delivery and access platforms. Broadcasting the TV content in digital form provides at least 6 times capacity savings on the

allocated frequency bands of the delivery medium whether it is air, cable or satellite. Since these bands are essentially natural resources for a country, in the past a few years, the governments of most European countries and USA, have declared that Analog TV broadcasting will be switched-off with deadlines ranging from 2010 to 2015. The indications of this is enormous : a global digital delivery medium to every TV set will be established within a decade! 234.6 million households with a TV set, only in Europe, may give an idea of the size of the market. Besides, an estimated 40% of these houses receive more than one means of delivery (e.g. both satellite and cable) which means they can be served with digital content by more than one operator. Forrester Research predicts that Interactive TV (iTV) will overtake internet as Europe's primary e-commerce platform, and the subscriber base will be 80 million by 2005 (which was 18 million at the end of 1999).

Case Study : United Kingdom

For the last 5 years UK almost turned into a test laboratory for all the interactive TV ideas in the world. It is the only country where iTV is delivered using all the three known broadcast mediums :

1. Satellite
2. Cable
3. Terrestrial (air)

BSkyB who is the satellite iTV operator, enjoys its first to market advantage with a market cap of 50 billion$ as of March2000. However, the terrestrial operator ONdigital and the major cable operators NTL and Telewest are not lagging far behind. All these operators are trying to aggregate service providers and merchants on their TV commerce feature, but again the highest return is on “Open” which is the shopping service platform owned 80% by BskyB. Latest news on the company was the purchase of %35 of Open shares (worth of 590 million$) from HSBC bank in August2000. HSBC bank who is known to have invested 100 million$ when Open had been established a year ago, showed once again the leading role of financial institutes in developing technology.

Open which is launched in October2000, was reported to have 750,000 email subscribers and 400,000 regular shoppers as of July2000. Domino Pizza, who is one of the 140 merchants (as of July2000) on Open, claimed to have sales of 40,000 pizzas per week on Open.

Abbey National (Britain’s 6th largest bank) started in October2000 to serve on Open together with HSBC. They were the first. In autumn 2000 there will be more than 20 financial institution serving over UK digitalTV. Abbey National itself have started to serve over all the digitalTV platforms in July and it announced that it expects 4 million homes to have TV banking available by the end of 2000 and 9 million by 2003.

Digital TV commerce is catching on in other European countries as well. Particularly in Holland, France and Spain.

4.2 Banking Products

4.2.1 Traditional products

Essentially, your 'brick & mortar' bank offers you three types of services:

1. Transactional : Clearing your checks, paying your bills, processing direct payments.
2. Credit : Providing you with credit, either through mortgages, credit cards or basic loans.

3. Investments : Providing you with investment options, including advice.

But you would sometimes have to make an appointment, or worse stand in a long queue. Electronic delivery channels provide all these plus more!

4.2.2 The new banking products

The cost reduction of the new channels together with increasingly many processes being automated and computerized in the financial industry have let many new banking products to emerge :

1. Free checks : Online banks now offer you free personalized checks, mailed to your home.
2. Free stock trading : Fierce competition among the banks has prompted major powerful banks to offer free stock trading.
3. Free Electronic Fund Transfers (EFT) : All online banks offer you the ability to transfer money between accounts, or even between banks for free!
4. Free online bill payment : Pay your utility, phone, credit card and other bills electronically without standing in a queue.
5. Export financial data : One of the forthcoming services, especially for the small or medium sized enterprises (SMEs) will be the transfer of their bank statements electronically to their financial software.
6. Instant mortgage approvals : Online application to obtain mortgage loans within minutes will be among the next set of services.
7. Instant credit card applications : Applying for credit cards online to get a decision within minutes.
8. Buy CDs online : Not the music variety, but the Certificates of Deposit (CD). Soon it will even be possible to price the rate of interest you would prefer on your CD.
9. Refer to past transactions : Online banks allow you to refer to past transactions (committed whether online or offline), even up to many months online.
10. Free interactive help : Online banks can provide resources to help you plan a budget, buy a new home and many other financial decisions.
11. Your own financial community : Banks offering brokerage services will provide their customers to contact like-minded others. Exchanging gossip can be profitable!

4.2.3 Segmentation

Electronic delivery channels act upon digital data which is inherently portable and observable. Currently it is an active project for every major bank with electronic channels, to aggregate this data in data-marts to mine for information about general trends in their business.

Secondly, this ability to capture and analyze information, that has not been historically available, can be used to prepare personalized user interfaces to the customers. Bank marketers, another group that will see significant expansion as the banking business models change, will compete to use this kind of information as a means to more targeted and effective marketing initiatives.

5 Banking Systems for Electronic Delivery

Ever since computers are manufactured mainstream, banks have been using them for their core transaction processing needs. The practice was to run batch transactions overnight and use the computing power for realtime transaction requests daytime. Aside from bank's own batches of transactions, all the demand for

transactions were received through bank branches. Lately tough, this computing capacity is being shared by the alternative channels of the bank, all of which are electronic and hence can be consolidated to use a centralized common **back-end**.

These requests for transactions are collected in various forms and a so called **middleware** became imperative to translate the request into a form that the transaction engine would understand.

To complete the picture, the user interfaces that the customers interfere with to place their requests, together with interfaces that call center agents use, constitute the **front-end**.

A second perspective to comprehend the three-tier architecture is the business approach :

"The consumption of a financial product, is initiated by a business encounter, which results in a number of business transactions, set by the business rules".

Business encounters are events in which the customer comes in contact with the delivery channel (e.g., walking up to an ATM). A single business encounter could result in multiple business transactions. In the ATM example, a funds transfer and a withdrawal would be two business transaction in one business encounter.

Business rules that define the functional capability of the channel are programs that define the participants and the business processes used to execute a particular transaction. For example, if a customer makes a transfer funds request, the business rule defines what a transfer is and how the system must check the existing balance against the quantity of the transfer before subtracting the amount and relocating it.

Business transactions are the individual banking transactions that comprise a business function fulfilled by a number of computer transactions. An ATM funds transfer is one business transaction that requires five separate computer transactions--PIN authentication, balance verification, debit, credit and customer receipt.

The technical requirements for this three-tier architecture can be listed as follows :

1. Providing a Robust and Scalable Architecture : The successful infrastructure in the new banking environment will need to provide a highly robust and scalable architecture, with the ability to provide at least 6-fold growth in the short term.
2. Absorbing Sudden Change : IT infrastructures must have the ability to absorb additional significant growth from mergers and acquisitions.
3. Emphasizing Speed and Availability : IT infrastructures must be able to offer a response time of less than three seconds as seen from the customer, decreasing to less than one second by the end of the decade. Overall customer availability will be 99.5 percent, increasing to 99.9 percent by the end of the decade.
4. Shifting Focus : Directors must shift their focus from branch systems to virtual systems, and build a single real-time customer-related view of data across the entire bank.

5.1 Front-end and Presentation

The front end, otherwise known as the user interface, is the core of the user experience, as it is what customers use to authenticate themselves and initiate their orders for financial products. The first of one of the electronic front-ends was the ATM. The latest ones are internet and TV banking. For the telephony channel, the real front-end is the telephony handset, the IVR and the live call-center agent.

A popular paradigm for the banks is to use the same presentation at all their front-ends but with minor adaptations. Assume that the call center agents use the same HTML interface as the internet customers use but with additional capabilities or that the TV bank is the low resolution graphics version of the same web page or the NT computer in the ATM runs Windows operating system and a publicly available browser with a disguise.

The unified HTML front-end solution gains its popularity by the fact that each additional product would be reflected to all the channels at the same time. In cases where the bank prefers to develop different sets of products to be delivered from each channel, it can still keep some of them common. A well-known practice is to provide both the call-center agents and the internet customers with the same HTML presentation pages.

Among the electronic channels internet is the one which provides the bank with at least three main technology models for developing front ends:

1. Client-side browser access to the Web server: This option is a simple Web site model, whereby the financial institution builds a Web site that is supported either in-house or through an outsourcer. Consumers simply log onto the Web site and navigate through the content and the transaction services with a simple Web browser, such as Netscape Navigator or Microsoft Internet Explorer, resident on their PC.
2. Client-side thin client access to the Web server: This model allows customers to download a thin client software product to their hard drive to access the Web site and store data locally. The thin client software resident on the consumer's PC has buttons corresponding to Internet functions, such as "pay bills" or "check accounts." When customers request a transaction that requires updated information, the thin client software launches Navigator or MS Internet Explorer from the desktop, which brings them to the Web site of the institution to execute transactions. The thin client software is branded by the bank, so users' experiences are consistent whether they are connected to the bank's Web site or simply looking at account data in an offline mode.
3. Client-side thick client access to the Web server: This model is used to support access with personal financial management (PFM) products. PFMs are commonly used products in USA. Major vendors are Quicken and MS Money. As with the thin client model, the thick client can be branded to the bank site in the transaction-oriented sections and can launch Navigator or Explorer to allow the customer to access the Web site and download information into the PFM.

It is important to note that these models are not mutually exclusive of each other. Many financial institutions that support PFM access to their Internet financial services functions also support a browser-based access method.

5.2 Middleware and Messaging

The second technology component in the three-tier architecture is the middleware, in other words the transaction platform. In general, it serves two main functions:

1. Definition of the functional capabilities of the channel.
2. Definition of the interfaces, or integration, between the front-end applications and the back-end processors, (be them core processing systems, remittance processors, payment processors, presentment aggregators or product specific applications)

The middleware software, which acts as an application intermediator, enables communication between disparate applications, between an application and a database, or between disparate databases. Middleware facilitates the communication by receiving information in the language or communication method of one application and translating it into something understandable by the other.

Each application can be controlled by a set of commands called Application Programming Interface (API). The API of an application can be used to integrated it with other applications either directly (the way that PowerPoint graphs can be pasted into a Microsoft Word document), or by the middleware. APIs can be

bought from the vendor of the application or an already purchased API of another application can be used if the vendors of the disparate applications are in partnership and have agreed to share APIs to facilitate communication between their applications.

It must be considered that the banking environment consists of many applications and the middleware can be overloaded with software code to manipulate each of these by using separate APIs. Fortunately, a basic trend in the computer industry is to use messages between applications that would trigger right events in the application to fulfill the desired command. Being a strong alternative to direct commanding with APIs, messaging standards can successfully integrate applications that are developed by separate vendors. Messaging Integration through a common messaging standard essentially means that the applications use the same messaging protocol to interface with each other and pass information and requests back and forth. The protocol defines the method of engagement and transfer of information between participants on a network. An example of a financial services system that interfaces on the protocol level is the use of the standard ISO 8584 Automated Teller Machine network standard for integration with bank core processing systems. Another commonly used middleware product in the banking industry is the IBM MQSeries messaging platform.

In the recent years a number of reasons have multiplied the functions required from the middleware :

1. Increasing number of electronic delivery channels
2. Increasing number of financial products
3. Frequent mergers and acquisitions in the financial industry

Financial institutions who use additional APIs, every time a new product had to be integrated or those who distributed their applications along remote nodes of the enterprise using Java, have started to suffer from the “spaghetti syndrome”. The question was how to bring together all the different digital information flows and hold them there. The natural answer was to evolve the middleware into a fat pipe where messages from all the applications were carried to and from the back-end seamlessly.

5.3 Back-end and the Business Rules

For a financial institution to allow a particular transaction to be executed, such as bill payment or a stock trade, there must be a business rule associated with it to define it. Traditional banks that decide to make use of electronic delivery channels, can do the developments in-house and they can leverage their existing business rules. There are two main ways of architecturally structuring the location of the business rules:

1. Business rules accessed off the back end: With this method, for example the Internet transactional business rules are maintained on the back end and accessed, when needed, by the transactional platform (middleware) through defined interfaces. This method is most common with an in-house developed system where an object request broker or terminal emulation product is integrated to make calls to engage the business rules on the back end.
2. Business rules maintained in the middle tier: With this method, for example the Internet transactional business rules are maintained on the middle tier within the transaction platform. This is most common with packaged application vendors or outsourcers in which the vendor has defined the Internet financial services functions within its application and integrates to the back end only to retrieve information and to update accounts.

Whether business rules are considered to be a part of it or not, back-end surely requires a potent computing power. This power is mainly consumed by updating tables of databases with an enormous frequency. The total aggregated transaction requests and the accompanying data is constantly fed to the back-end computing facilities to be executed realtime or batch. For such computers used as transacting engines, it is clear that Input/Output speed is more imperative than the calculation power.

It is expected that in short term the required power for the back-end will increase significantly because of two major accelerators:

1. More computer transactions per business transaction : Convenience for the customer means more complex business transactions, which result in more computer transactions to complete a single business encounter
2. More compute power per computer transaction : Each computer transaction requires more compute power because of the migration from batch to on-line transactions, richer functional design to improve ease of use, additional security measures such as encryption, and use of advanced programming techniques such as object-oriented design.

Accompanying the computational concentration the back-end invariably includes three facilities :

1. Storage : An array of high speed disk drives must be attached to augment the memory in the transacting computing system.
2. Back-up : The state and the recent data processed by the transacting system must be backed-up on digital tapes with frequent enough intervals. Institutions of middle or large size, usually chose to install automated tape libraries which utilize robotic arms and can be managed/controlled by the same system that controls the transactions.
3. Disaster Recovery : A major core competence of banks is the reliability. This customer expectation enforces banks to take precautions against all kinds of disasters including natural catastrophes. Common practice is to maintain or outsource the replica of the whole back-end system together with the backed-up tapes. IBM, whose back-end systems are widely used in the banking world, provides this service for all the banks that have deployed its products and would chose to outsource the disaster recovery precautions.

5.4 Security

Security is the most important core competence of the financial institutions. These institutions are supposed to be the ones that would menage the risk for people who would like to invest or deposit their money. With the acceptance of electronic mediums to transfer the location of currency, security started to mean much more that just strong building constructions and safe-deposit boxes.

Each electronic delivery channel comes with its own concerns of security. Since most segments of these channels are used to transmit digital data, the kinds of attacks are rather technical then by force! When considering the security of the channels two components must be identified :

1) The Network : The attacks on the network is generally in the form of tapping-in and stealing data. The precaution is to use strong enough encryption of the data as it is being transferred over the network.
2) The Facility : The attacks on the banking facility could be either to disable its functions (Denial of Service – DoS attacks) or to login as an intruder for committing transactions or to spreading viruses to corrupt. The precaution is to use firewalls at the point where the facility is connected to the network.

Aside from their core businesses, banks and brokerages have helped lead the e-business and e-commerce charges over the past few years. This growth has also left them more vulnerable to a broader range of threats and uncertainties. Consequently, spending on IT security in financial services has ballooned in recent years. IDC reports that, worldwide security-vendor revenues in the finance/insurance sector grew from 2.4 fold every year for the last 5. HSBC Securities, the brokerage unit of HSBC Holdings PLC of London, has deployed a product called SurfinGate at a cost of around $100 per user.

5.4.1 Encryption

Encryption is the process of converting information into a more secure format for transmission. In other words the plain text is converted to scrambled code while being transmitted, and then decrypted back to

plain text at the receiving end of the transmission. It is comparable to writing a letter, converting it to code, putting it in an envelope and mailing it with the recipient descrambling the code.

For encryption to work properly, both the sender and receiver have to know what rule, or cipher, was used to transform the original information into its coded form, often called cipher text. A simple cipher might be to shift all characters in a message by an arbitrary number of characters, say 5. As long as the receiver knows that's what the sender did to the message, the receiver will subtract 5 characters from the message received to extract the original text. (It is believed that this particular type of cipher was used by Julius Caesar to communicate with his commanders.)

Currently, there are 2 levels of encryption generally available in web browsers: 40-bit encryption, and 128-bit encryption. Most commonly available internet browsers use 128-bit encryption. Handheld devices or smartcards usually provide the equivalents of 40-bit encryption. The difference between these two types of encryption is one of capability. 128-bit encryption is exponentially more powerful than 40-bit encryption.

This encrypted channel between the customer and the institution is established just above the physical layer of the network and hence called Secure Socket layer (SSL).

There are 3 ways to determine if the site on which you're banking is secure :

1. Go to your bank's homepage and log into your account by inputting your user-id and password. Notice that the lettering in the URL address will change from "http" to "https." The extra "s" stands for "security," and lets you know that the page is indeed secure.
2. At the bottom left corner of your screen, a padlock icon will appear.
3. You may also find a gold seal somewhere on the site that reads, "VeriSign Secure Site; Click to Verify." VeriSign is a leading provider of Internet trust services, and its insignia is confirmation that your bank is certified as a secure domain.

5.4.2 Firewalls

DoS attacks are just one of many invasion threats that banks and brokerages face. To counteract them, institutions most commonly use firewalls, intrusion detectors like anti-virus software (also called malicious code detectors) and vulnerability scanners.

Firewalls are installed between the institution's connection with the Internet and its own networks. They are configured either by the user or the firewall itself to filter traffic passing into or out of the internal networks. They can, for example, prevent programs from being downloaded off the Internet or block some kinds of departing email.

Intrusion detectors identify and deactivate "malicious codes," or unwelcome online guests such as viruses, so-called Trojan Horses, and Java applets and ActiveX controls. Trojan Horses enter systems because they are in the form of something that firewalls have been pre-configured to admit-generally an email message. Java applets and Microsoft's competing version, ActiveX Controls, are known as "mobile code" because they are automatically downloaded off the Internet and executed along with the application they are attached to. This makes them difficult to be filtered out of a network.

In the case of banks and brokerages additional security means may also be necessary like firewalls enhanced with intrusion-detection software. Integration of such additional software typically occurs at the firewall level. Symantec recently developed Norton Antivirus for Firewalls 1.5, which can be installed on IBM- and CVP-compliant firewalls. Computer Associates markets firewalls integrated with content filters and malicious-code scanners that can be instantly upgraded when new invaders appear.

Unfortunately, most malicious-code detectors can only eliminate intruders that are already known. That's a problem because a new fast-spreading code like that based on Java applets and ActiveX controls can do enormous damage before "antibodies" can be generated.

A relatively new line of products against mobile malicious-code, resides on the firewall and protects networks from all mobile code. It scans incoming traffic for mobile code, isolates it and tests the application it will run in a pre-configured "sandbox." If the application is outside pre-set parameters, the entry is blocked.

This would prevent, for example, a bank's network from being breached by a brand-new piece of malicious mobile code, such as last year's Melissa email virus. Such prevention could save a bank both from having its own and customers' data compromised and from incurring downtime expenses.

5.4.3 Authentication

More and more companies are conducting high-value transactions with their banks and brokerages over the Internet, as well as conducting business with other companies through their financial services providers online. That has pushed banks and brokerages to invest more heavily than other industries in cutting-edge access contro technologies. Most forms of network access control involve some form of "user-authentication" technologies. These allow both the sender and recipient of online transactions to verify themselves and each other, most commonly by attaching a "digital certificate" or some other form of digital signature, which could even be an electronic fingerprint.

The necessary infrastructure for user authentication, known as public key infrastructure, or PKI, consists of two security "keys." One is the public key (the digital certificate), which is either downloaded from a browser or called up from a hard drive and attached to the transaction command or document. By ratifying the sender of the message as the subject of the request, it acts as a kind of digital driver's license that identifies the sender of a transaction and provides certain details about him.

The other component is the private key, also called the RSA key, after RSA Security Inc., the company whose subsidiary is the market leader in digital certificate supply. The private key sits on the hard drive of a computer and, when activated by a user name and password, unlocks the user's PKI access.

On the other hand, some bankers and security analysts point out that the PKI-digital certificate system has security weaknesses. They say the private key can still be reached by a clever hacker over the Internet, so sealing entry to the private key with "just" a user name and password is no longer considered secure enough. Security providers have developed a variety of solutions to this problem, collectively known as "extended user-authentication." Essentially, these technologies, which can be hardware- or software-based, require the user to enter some form of secured identification to access the password or the private key.

Tokens, also called "keys to the key," are external hardware devices plugged into the computer. They generate a code that the user must enter into the system to unlock access to their private key. The device adds to security by changing this code each time it is used.

Theoretically, to establish complete authentication three means are enough :

1. Something the user knows
2. Something the user has
3. Something the user is

The third means is the study of the science called biometrics. An example application of biometrics is an external device, such as a smart card, that carries an authentication biometrics, such as a fingerprint. Fingerprint imaging, like other biometrics such as iris-scanning and signature-reading, converts the different recurring patterns in unique human physical properties into a digital code to be stored, using special algorithms.

6 Banking Organization for Electronic Delivery

The functional approach challenges bankers to determine how they can most efficiently perform the various financial functions their customers want. In today's financial markets, where currency can find many ways of multiplying itself or be transferred, what is the comparative advantage of banks? Another way of addressing this question is to think about what functions can a bank profitably outsource. Money market mutual funds, for example, outsource some risk management. By engaging in loan sales and securitization, a banking firm can outsource the transfer and pooling functions, but keep the credit evaluation and monitoring (for example, risk management and incentive) functions.

In functional thinking, the initial tendency is to assume that everything can get outsourced, broken up, and reduced to a commodity. The main argument for outsourcing is that the bank's staff needs all the help it can get in dealing with new technologies. The counter argument is that outsourcing reduces the bank's control over its services. A "best-of-both-worlds" solution brings in the experts at the start of a program. Then, when staffers get the hang of the new technology, they can take over.

Clearly for taking any of the sides, to identify the costs is the main parameter. One obstacle for evaluating costs of engagements is that most banks have established their utilization of electronic channels by a joint effort of their various existing departments. This was a short term solution as the channels were evolving fast. Having no centralized operation, prevents the banks from scaling up their resources for increased demand. It also makes it hard for them to identify the cost of the endeavor and the operation.

This is the reason why many financial companies are restructuring at the moment to establish their electronic delivery channels with related functions, as separate business units and cost centers.

7 Future Trends

The competitive landscape for retail banking services in the future seems to have four prominent features :

7.1 Breakdown of Geographic Barriers

If the consumer (or small-business manager) can conduct the full range of banking business with a telephone, personal computer, or an ATM, will it matter as much in the future where the physical location of the bank is? In other words, geographic convenience could make little difference in the future. In which case, the consumer will choose banking services largely on the basis of price considerations -- as many consumers already do for credit cards. With banks advertising their services and accepting applications electronically over the Internet, a time may come in the not too distant future when the consumer will be able to conduct a large amount of highly automated "comparison shopping," using so-called "intelligent-agent" software programs to search for and look at products from all accessible banks, not just from those banks that happen to be located near the consumer's home or work place.

7.2 Commoditization of Banking Products and New Competitors

Customers will not necessarily look for a "complete bundle" of banking services from a single bank, but could bank on a product-by-product basis, or, as financial products become increasingly standardized as they become automated, perhaps on a "commodity-by-commodity" basis. That is, electronic delivery of banking services could erode the brand-name loyalty consumers now have to their banks. Moreover, for products like credit cards, mortgages, auto loans and some investment and savings products, there will also be non-bank competitors, perhaps "category killers" specializing in being a very large provider of just one product to gain the lowest unit costs through economies of scale in processing. The advantage of these nonbank competitors may also stem from the lack of the large fixed cost of an extensive branch network. They can operate largely through telephone, mail, and eventually perhaps Internet distribution.

7.3 New Competitive Strategies for Banks

Some banks may choose to unite their branch distributions with other retailers such as supermarket chains by using a small but high-tech branch structure. These scaled-down, but strategically placed, branches might help these banks capture that segment of the retail market that is not ready for total electronic delivery of financial products, or those who might not actively seek information about financial products.

Banking kiosks is just another version of this strategy. A bank may open kiosks across the country in the offices of a partnering travel-service company, a rather bold move to gain a national presence at low cost.

7.4 Banking on Technology

Banks are likely to keep their leading role as fast technology adopters in the years to come.

7.4.1 Customer relationship management - CRM

Within the next few years, you can expect call centers in the finance and banking industry to continue automating their customer service procedures. Most banks, now have an IVR system which has been evolving many years and could handle upto 80% of the incoming traffic. Although they will rely more on IVR systems and on e-mail than in the past, there is no great danger of fewer job opportunities or layoffs for agents. Technical advances aren't intended to make live phone service obsolete, but rather to enable agents to work more efficiently. Customers don't always mind working with an IVR system's menu-driven format or waiting for an e-mail response for a simple matter like requesting an account balance, and they are more likely to speak to agents only about more complex transactions.

Speech recognition is also a mature technology now, that the call centers can readily make use of. It is available for many languages. In fact the advancements in speech recognition is mostly on porting the existing systems on other languages.

Other features of speech technology currently being developed are :

1. Voice recognition : Vocal models of customers can be collected and checked in a manner which would be transparent to the user. Especially for the telephony channel, voice recognition can be a very effective auxiliary authentication method.
2. Natural and continuous speech recognition : Regarded as the ultimate achievement in the speech technology, computer recognition of human language would require advancements in the recognition of speech in continuous form. This would give way to more natural interactions with the customer. On the other hand, as the natural language understanding technology evolves not only telephony agents would benefit, but also Email automation facilities would be greatly improved. Natural language processing abilities are considered to be great steps in evolving call centers into contact centers.

7.4.2 Virtual banks

Generally, the future will rely more on people and technology, and less on physical location. People will phone or e-mail an expert rather than talk face-to-face with a local teller, especially in areas where expertise is vital: life insurance, financial planning, and mutual funds. But Internet-only subsidiaries first must overcome many hurdles. Having conscious customers is one thing; establishing "core" online relationships is quite another. Cyber Dialogue, an Internet database marketing company, estimates that almost a third of the 9.4 million people who have signed up for online banking have discontinued the service and says only 35% of the defectors are inclined to try it again. The problem was overly complicated technology and inadequate levels of customer service. In the United Kingdom, for example, two established players have set up Internet companies with catchy names Egg from Prudential Banking PLC, and Smile from Co-operative Bank. They are competing fiercely for customers, against First Direct and Virgin Direct, both of which started out by conducting transactions over the telephone and now have added Internet access. Egg has managed to make a bigger impression on the U.K. market than Bank One's Wingspan has in the United

States. Wingspan has attracted about 50,000 customers over about six months. Egg, with more than $12 billion of deposits and 600,000 customers, celebrated its first anniversary in October1999. These numbers make Egg six times bigger than Telebank, a U.S. branchless bank that has been on the Internet for four years and has $2 billion of deposits and 100,000 customers. Egg has set itself a target of two million Internet customers by 2004.

In USA, web users can gain access to these services at some well-known sites such as CNBC.com and Ameritrade.com, but the bulk of the data-gathering, record-keeping, and security functions belong to one of two companies : VerticalOne Corp.(a subsidiary of bank adviser S1) and Yodlee Inc. Both data processors, or "aggregators," are little more than a year old. Unlike banks and brokers, they are unregulated and may have no responsibility for fraud. The two are quietly amassing huge amounts of valuable data on consumers' buying, borrowing, payment, and investment habits.

Here's how the aggregation process works : A customer signs up for the finance-managing service at a "partner" Web site such as CNBC.com. In setting up the account, customers must type in their account numbers and passwords--information that goes directly to VerticalOne or Yodlee. Every night, their computers go online to Web sites run by the banks and brokers, and log on with the customer's name, account number, and password to retrieve information. This data-gathering process is known as "screen scraping."

The account information is separated into names, numbers, and passwords. It's then encoded and stored in high-security databases that use palm-scanners for access. Although together the companies have several hundred employees, there are only six (three executives at VerticalOne and three at Yodlee) who have complete access to the information. VerticalOne and Yodlee insist their security is bulletproof.

7.4.3 Data mining

Banks are also beginning to realize that they have a large amount of information about their customers in their electronic data bases and are making an effort to consolidate this information into data warehouses so that they can target certain customer segments for new product offerings and not waste resources making the same product offering to other segments that probably would not be interested. The transmission of the product offerings to the targeted consumer segments will become more electronic, with banks "pushing out" the information over the ATM and personal computer networks. Personalized messages will appear on the computer or ATM screens when the targeted consumers access these networks for their routine banking business. Information consolidation will lead to cost reduction as well.

Banks can use the information from deposits to better price and monitor loans. Because it is the intense information uniqueness that prevents illiquid loans, such as small business loans, from being customized and securitized, this added informational advantage is decisive.

7.4.4 Intelligent agents

Intelligent agents (IAs)are software applications with specific goals which are delegated to them by their user. Their other properties are being autonomous and communicative. Once their purpose is established and operational parameters are set, IAs have control over their own actions and do their work in a self-perpetuating, self-monitoring fashion. IAs are reactive they can sense changes in their environment and respond accordingly. They can communicate with each other and relevant systems that constitute their surrounding. The general activity areas for IAs can be listed as follows :

- IAs can speed the retrieval of customer related information by rapidly searching multiple data sources, making them especially well-suited for troubleshooting assistance or ad hoc queries. The result is fast and more effective customer service.

- IAs can boost Internet-based self-service effectiveness by sharing the expertise "learned" during interactions with other customers. As a result, customer searches grow continually more effective and efficient over time. IAs can be used by organizations to enhance their relationship with high-net-worth customers.
- IAs can assist in data-mining activities by searching tirelessly for trends in customer data, even if the data resides in multiple data stores. As a result, data relationships can be discovered that might otherwise go unnoticed by human eyes.

7.4.5 Biometrics

Traditionally banks have relied on very simple forms of identification such as passwords or personal identification numbers for banking transactions. These methods are inaccurate and provide an opportunity for unauthorized individuals to access banking accounts. However, vendors are producing new devices that use biometrics technology for banking security. Currently almost all banks are investigating some form of biometrics for their alternative, electronic delivery channels. This technology, though mature enough, is not yet cost effective for the banks. However with the ever dropping prices of silicon chips, within a few years various biometrics measures are expected to be commonplace in the banking services.

Delivery Channel:	Key Function:	Biometrics For Consideration:
PC/Internet banking	Verify identify of banking customer for PC or Internet transactions	Fingerprint, face or voice
ATM	Verify identity of ATM user	Fingerprint, retina, iris, face and voice
Telephone banking	Verify identity of banking customer for telephone transactions	Voice
Branch	Verify identity of individuals cashing checks without banking accounts Verify identity of banking personnel for internal computer security Verify identity of banking personnel and customer for physical access security	Fingerprint Fingerprint, voice, face, retina, and iris Fingerprint, hand, iris, retina, face, finger geometry and voice

Table 3 Key Functions for Biometrics in Retail Banking Product and Service Distribution

7.4.6 Mobile wireless access devices

Two converging trends are driving the financial industry's interest in supporting wireless device access by their consumer customers. The first is the development of the Internet as a channel for financial service delivery, and the second is the proliferation of cell phones, PDAs, pagers and other unwired access devices throughout the consumer market. The estimated worldwide shipments of handheld units in 1998 was 4 million. Indicators of consumer adoption of the Internet for financial services could be given as the number of U.S. consumers trading stock over the Internet being 4 million, and the number of U.S. consumers using the Internet for banking being 5.6 million as of early 1999. Considering that both device proliferation and Internet financial services adoption are still in strong growth stages there is reason to believe that these trends will overlap, particularly since the work force is increasingly mobile and in need of convenient ways to manage everyday financial needs from remote locations. In addition, the hype surrounding the mobile markets is likely to continue with the emergence of 3rd generation mobile networks and mobile devices with excessive computation and multimedia capabilities. GartnerGroup, a technology analyst company, predicts

that at least 25 percent of all consumers that use the Internet to manage their finances, will also own a wireless data-enabled device by year-end 2004.

8 Conclusion

Technological innovations have dramatically changed retail banking in the world. Many banking business functions, from product development to product servicing, have been automated. However, technological advancements are and will continue to make the greatest impact on retail banking distribution, including both making traditional channels, such as branches and ATMs, more efficient and supporting the emergence of new channels, such as the Internet and call centers. As banks move into the twenty-first century, new technology development will continue to mold the retail banking business. Technology will make it a more frictionless world for money.

9 Resources

Bank Administration Institute (BAI) Online : http://www.bai.org/index.html

A financial portal : http://www.clearlybusiness.com/default.htm

American Bankers' Association (ABA) : http://www.banking.com/aba/default_0700.asp

Banking in the On-Line World : http://www.rbrldn.demon.co.uk/online.htm

American Banker : http://www.americanbanker.com

The Financial Services Roundtable: http://www.fsround.org/

Technology Group for The Financial Services Roundtable: http://www.bitsinfo.org/

Links list of Independent Community Bankers of America : http://www.ibaa.org/links/links_pubfr.html

The Electronic Payments Association : http://www.nacha.org/

Banking Technology Magazine: http://www.bankingtech.com/

An article on Banking History : http://www.drennangroup.com/History/history.html

Retail Banking Taxonomy : http://www.towergroup.com/pages/retail.asp

A report on the new economy : http://www.esa.doc.gov/de2k.htm

A free publications from Institute for international economics : http://www.iie.com/publications/publication.cfm?pub_id=318

A 1999 Report by the Federal Bank of Dallas : http://www.dallasfed.org/htm/pubs/annual/arpt99.html